

Monday 08/31/2026: Theorems

CSCI 246: Discrete Structures

Textbook reference: Sec. 4, Scheinerman

Logistical Matters

- **Reminder:** Your two jobs before each class are to (a) make sure you can do all the group exercises and (b) read the next section (see syllabus).
- **Office Hours:** My and Madhav's office hours have been posted to the syllabus.
- **Weekly Quiz:** This Friday's weekly quiz will have three questions, and will cover definitions and theorems.

Weekly quiz

The Friday weekly quizzes will cover group exercises and the readings from a batch of three class meetings (that week's Mon., and the previous Wed. and Fri.)

Today's Agenda

- Mini lecture ($\approx$ 35 mins)
- Group exercises ($\approx$ 15 mins)

Motivation

Logic

If-then statements

If-and-only-if statements

The big picture: propositional logic

Why care about theorems?

Shrinking data without losing distances

Applied Problem. Every Netflix user is a vector of **50,000** movie ratings. To find similar users, *each* distance you compute costs 50,000 operations. **Can you shrink the vectors and still find similar users?**



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Possible Solutions.

- *Drop coordinates?*

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- *Drop coordinates?* Two users differing **only** in the dropped ones now look identical.
- *Keep the most informative ones?*

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Possible Solutions.

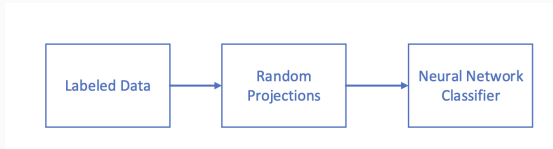
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Remark. This solution doesn't require looking at the data **at all**!

Why care about theorems?

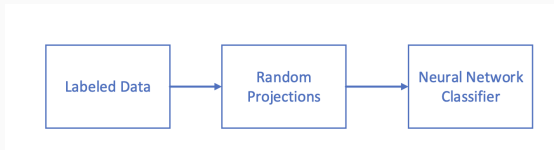
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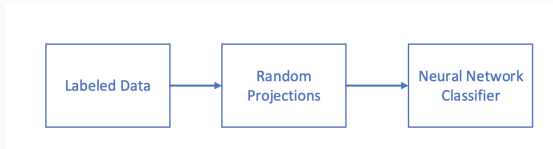
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Motivation

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If-and-only-if statements

The big picture: propositional logic

Logic

The statements in a theorem are written in a **formal language**. Its vocabulary is tiny: a handful of **logical connectives**:

and	$\wedge$
or	$\vee$
not	$\neg$
if-then	$\implies$
if and only if	$\iff$

Also called **logical operators**, the same things your programming language spells `&&`, `||`, `!`.



Logic (the rapper)

Practice Quiz on Reading

Replace each ? with a checkmark ✓ if the combination of truth values for propositions A and B is *possible* under the given logical connective.

Replace it with a ✗ if the combination is *impossible*.

Propositions		Logical connective		
A	B	if A then B	if B then A	A if and only if B
T	T	?	?	?
T	F	?	?	?
F	T	?	?	?
F	F	?	?	?

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The goal of today's slides is to clarify the solution.

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If-then: Truth Table

Given two propositions A and B , the new proposition “If A , then B ” ($A \implies B$) can be written in terms of this *truth table*:

A	B	If A, then B
		$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

The green rows are **consistent** with the statement, and the red rows **violate** it.

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Example. Let A = A person lives in Bozeman, B = The person lives in Montana. Let's check this against the truth table, row by row.

If-then: Example

Let A = A person lives in Bozeman and B = The person lives in Montana.

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Let A = A person lives in Bozeman and B = The person lives in Montana.

A	B	If A, then B	
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T	T	T	Me
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F	T	T	
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If-then: Example

Let A = A person lives in Bozeman and B = The person lives in Montana.

A	B	$\overbrace{A \implies B}^{\text{If A, then B}}$	Example
T	T	T	Me
T	F	F	Not possible!
F	T	T	
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If-then: Example

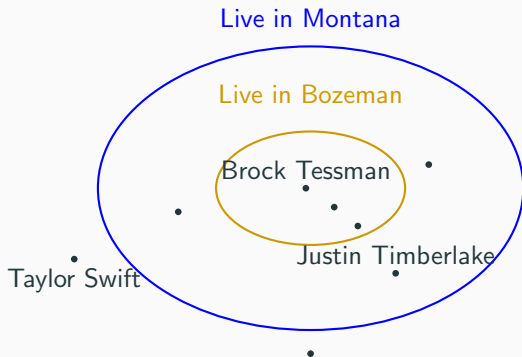
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Conclusion. The only row that would **violate** the rule (Bozeman but not Montana) **cannot happen**. Therefore $A \implies B$.

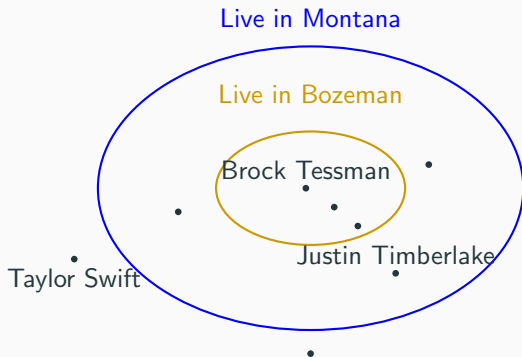
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Set theoretic perspective. You can think about $A \implies B$ as “the set of things satisfying property A is contained within the set of things satisfying property B .”

Poll: True or False? Every elephant flying with polka-dotted balloons above Bozeman is smoking a cigar.



Vacuous Truths

Solution to poll

Let A = this is an elephant flying with polka-dotted balloons above Bozeman and B = this is smoking a cigar. Then A is always False. So $A \implies B$ is **TRUE!!!** This is called a **vacuous truth**. Per Scheinerman, these statements are true “because they have no exceptions.”

Vacuous Truths

Solution to poll

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Reference material

Given two propositions A and B , the statement “If A , then B ” ($A \implies B$) holds whenever following “truth table” applies:

A	B	If A, then B $A \implies B$
T	T	T
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If-then: Valid argument and logical fallacy

Modus Ponens (A valid argument) ✓

Structure

If A, then B.

A.

Therefore B.

Example

If it is raining, the ground is wet.

It is raining.

Therefore, the ground is wet.

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Affirming the Consequent (A logical fallacy) ✗

Structure

If A, then B.

B.

Therefore A.

Example

If it is raining, the ground is wet.

The ground is wet.

Therefore, it is raining.

If-then: Other jargon

Some other terminology which means the same thing:

- A implies B
- A is sufficient for B
- B is necessary for A (A "only if" B)

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The big picture: propositional logic

If and only if: Definition and shorthand

If and only if statements: Definition

$\underbrace{A \iff B}_{\text{A if and only if B}}$ means $\underbrace{(A \implies B)}_{\text{if A then B}}$ and $\underbrace{(A \impliedby B)}_{\text{if B then A}}$

If and only if: Definition and shorthand

If and only if statements: Definition

$$\underbrace{A \iff B}_{\text{A if and only if B}} \quad \text{means} \quad \underbrace{(A \implies B)}_{\text{A only if B}} \quad \text{and} \quad \underbrace{(A \impliedby B)}_{\text{A if B}}$$

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Shorthand. For convenience, we can abbreviate A if and only if B as A iff B.

If and only if: Truth Table

The truth table for the “and” operator (also written $\wedge$) is given by

Original propositions		New proposition
X	Y	$X \wedge Y$
T	T	T
T	F	F
F	T	F
F	F	F

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F	T	F
F	F	F

Now we apply the $\wedge$ operator to the results of the $\implies$ and $\impliedby$ operators.

Orig. props.		New props.		
A	B	$\overbrace{A \implies B}^X$	$\overbrace{B \implies A}^Y$	$\overbrace{(A \implies B) \wedge (B \implies A)}^{X \wedge Y}$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

If and only if: Truth Table

Given two propositions A and B , the statement “A if and only if B” (that is, the property $A \iff B$) holds whenever following “truth table” applies:

A	B	A if and only if B
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If-then: Example

Let A = This month contains Labor Day and B = This month is September.

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If-then: Example

Let A = This month contains Labor Day and B = This month is September.

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T	T	T	September 2021
T	F	F	
F	T	F	
F	F	T	

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Let A = This month contains Labor Day and B = This month is September.

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F	T	F	Not possible!
F	F	T	April 2026

If-then: Example

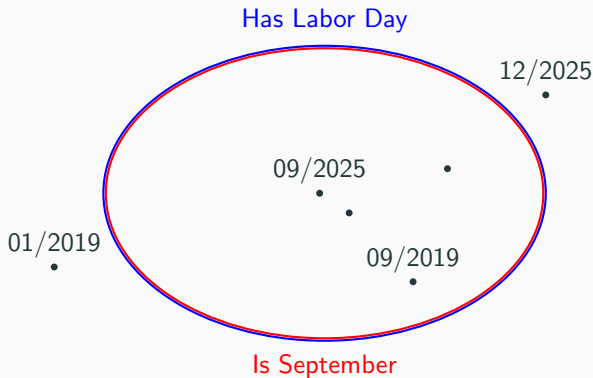
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Conclusion. The only rows that would **violate** the rule **cannot happen**.
Therefore $A \iff B$.

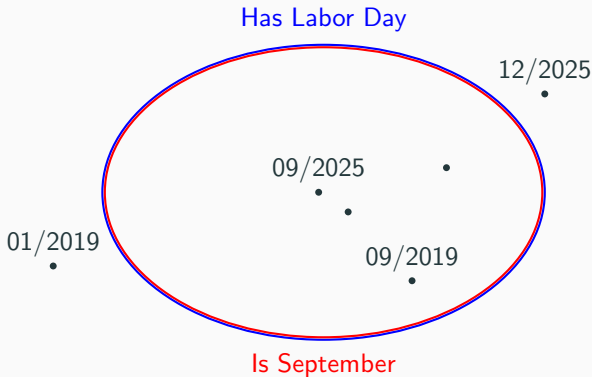
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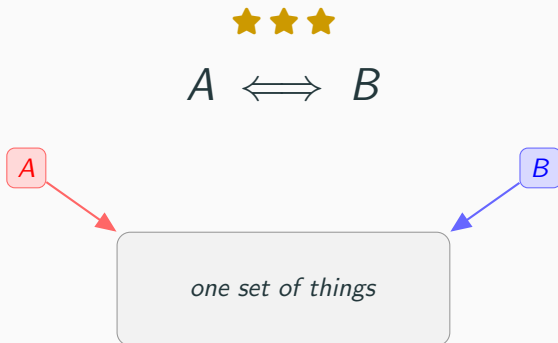
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If and only if: Big picture



A and B are two different **descriptions** of the **same thing**.

If and only if: Why should computer scientists care?

Consider the following theorem.

Theorem (Graph Theory)

A connected undirected graph with n vertices has a cycle if and only if it has at least n edges.

Thanks to the “iff”, you know that edge count alone is enough to decide whether a cycle exists. Instead of running a full DFS or Union-Find cycle detection algorithm, you can just check counts:

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1 def has_cycle(num_vertices, edges):  
2     return len(edges) >= num_vertices
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Take home. Why would computer scientists care about theorems of the form “A iff B”? It can sometimes happen that what we really want is A, but that’s very hard to code up. Thanks to the theorem, we can just code up B instead, because it turns out (perhaps surprisingly!) that it’s the exact same thing.

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Solution to practice quiz

A	B	if A then B	if B then A	A if and only if B
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
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Propositional logic

Original propositions		New propositions		
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Terminology. The first two columns combined with one remaining column gives the **truth table** for that logical operator.

Why are we doing all of this?

Objection. Why should computer scientists be concerned with formal structures (definitions, theorems, logic, etc.)?

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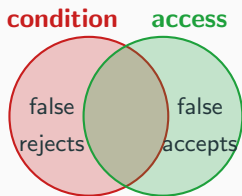
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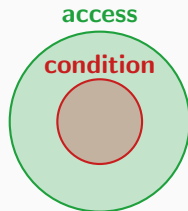
In general, the iff condition captures both soundness and completeness.

- Here, upgrading to an iff ensures no unintended backdoors or bypasses

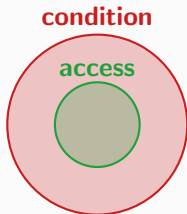
Soundness and Completeness of an Authentication System (Venn examples)



neither sound nor complete



complete (no false rejects) but
not sound



sound (no false accepts) but not
complete

condition / access



sound and complete

Group exercises

Random group assignments

Ashley, Isabelle: 5

Bailey, Tanna: 12

Blaisdell, Olyver: 2

Brown, Ashton: 8

Brown, Jonathan: 13

Cardenas, Dominic: 6

Cutler, George: 2

Fabik, Roland: 10

Fellows, Audrey: 7

Fitzgerald, Ryan: 12

Fuller, Ryan: 9

Griffen, Clara: 4

Grutsch, Tanner: 13

Hager, Nathan: 9

Horner, Jayden: 1

Johnson, Jaret: 7

Kintzel, Samantha: 1

Lameres, Alexis: 6

Lee, Holden: 6

Mayala, Reece: 10

McLean, Connor: 8

Mcbride, Daniel: 9

Miller, Maverick: 11

Moss, Emily: 5

Mousseau, Matthew: 11

Neuman, Evan: 7

Putnam, John: 3

Ramos, Vydel: 13

Rathbun, Jedidiah: 4

Roberts, Matthew: 11

Roller, Nathan: 10

Ryan, Otis: 3

Shevtsov, Timothy: 5

Sinclair, Aidan: 3

Smith, Charles: 4

Stensland, Emma: 2

Thiessen, Logan: 8

Thompson, Jaiden: 12

Vath, Ethan: 1

Group exercises

1. It is a common mistake to confuse the following two statements (i) If A, then B and (ii) If B, then A. Find two conditions A and B such that statement (i) is true but statement (ii) is false. Then find two conditions A and B such that both statements are true.
2. Consider these two statements: (i) If A, then B, (ii) If (not B), then (not A). Under what circumstances are these statements true? When are they false? Explain whether these statements are identical or not. [Note: (ii) is called the **contrapositive** of (i).]
3. (Challenge problem, from philosopher Norman Swartz.) Is the following statement true or false, and why? *A's-being-a-necessary-condition-for-B is both a necessary and sufficient condition for B's-being-a-sufficient-condition-for-A.*

Question 1: Solution

$A \implies B$ but $B \not\implies A$:

A = I lived in Los Angeles

B = I lived in California.

$A \iff B$:

A = Valentine's Day is this month

B = This month is February.

Question 2: Solution

Recall from Slide 7 that the truth table for the $\implies$ operator is given by

X	Y	If X, then Y $X \implies Y$
T	T	T
T	F	F
F	T	T
F	F	T

Now we apply the $\implies$ operator to the results of the "not" operator (also written $\neg$).

Orig. props.		New props.		
A	B	$\neg A$	$\neg B$	$\neg B \implies \neg A$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Note that $\neg B \implies \neg A$ gives the same results as $A \implies B$ as on Slide 7.

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T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Note that $\neg B \implies \neg A$ gives the same results as $A \implies B$ as on Slide 7.

Remark: We've shown that a proposition is logically equivalent to its contrapositive. So what? Sometimes it's easier to verify the contrapositive version.

Question 3: Solution

The simplest way to see this is as follows:

- A's-being-a-necessary-condition-for-B can be expressed as $B \implies A$.
- B's-being-a-sufficient-condition-for-A can be expressed as $B \implies A$.
- In other words, both propositions are the same: $B \implies A$. And a proposition is always necessary and sufficient for itself.